

Precursor

Cross-Asset Momentum Spillover from Commodities to Sector Equities via Granger Causality

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This notebook constitutes the full research manuscript. All figures and statistics are generated programmatically from the preceding analysis notebooks (01–06) and are reproducible from the raw data.

Abstract

This paper investigates whether momentum in commodity futures markets contains statistically significant predictive information about subsequent momentum in related sector equity ETFs. Using daily data from January 2010 to May 2026 across five commodity-equity pairs, we apply Granger causality testing, Vector Autoregression (VAR), and Directed Acyclic Graph (DAG) analysis to characterise the cross-asset causal structure. We find that all five hypothesised commodity-to-equity spillover relationships are confirmed at the 5% significance level, with Gold→GDX producing the strongest signal ($F=18.90$, $p<0.001$) and NatGas→XLE the weakest ($F=2.43$, $p=0.046$). Forecast Error Variance Decomposition reveals that commodity shocks explain 46.4% of GDX momentum variance and 31–37% of XLE momentum variance at short forecast horizons. Rolling Granger tests and Chow structural break analysis reveal that the causal relationships are regime-dependent rather than structurally persistent: near-universal structural breaks are detected at the 2022 energy crisis across all pairs, while COVID-era breaks affect energy and materials pairs but not Gold→GDX. A signal-based long/short backtest confirms the regime-conditionality of the spillover: strategies achieve positive Sharpe ratios during the COVID volatility regime (2020–2022) but deteriorate in the post-energy-crisis period. These findings suggest that commodity-to-equity momentum spillover is a genuine but time-varying causal phenomenon, activated by macro-structural shocks rather than operating as a continuous arbitrage.

1. Introduction

The relationship between commodity prices and the equities of firms that produce or consume those commodities is a well-documented stylised fact in financial economics. Oil prices influence energy company revenues; copper prices signal industrial demand that flows through to materials sector earnings; gold prices drive the profitability of gold mining operations. What is less understood is whether these relationships manifest as *predictive lead-lag dynamics* at the momentum level — that is, whether the directional trend in a commodity's price contains information about the future directional trend of a related equity sector, beyond what the equity sector's own price history already reflects.

This question has direct implications for both asset allocation and our understanding of how information diffuses across asset classes. If commodity momentum systematically precedes equity sector momentum, then commodities function as upstream information signals — their price trends reflecting fundamental supply-demand shifts before those shifts are fully priced into equity valuations. The lag between commodity price changes and equity price adjustments could arise from several mechanisms: the time required for commodity price changes to flow through to corporate earnings, the speed at which analyst forecasts are updated, or the rotation speed of institutional capital across asset classes.

This paper makes the following contributions:

1. We apply formal Granger causality testing to momentum signals (rather than raw returns) across five commodity-equity pairs, providing a more direct test of the momentum spillover hypothesis than prior correlation-based approaches.
2. We construct a Directed Acyclic Graph (DAG) of the full cross-asset causal network, computing centrality metrics that identify which commodities function as the strongest leading indicators and which equity sectors receive the most predictive signal.
3. We document the time-varying nature of the causal relationships through rolling Granger tests and Chow structural break analysis, showing that the spillover is regime-activated rather than persistently present.
4. We provide Impulse Response Functions (IRFs) and Forecast Error Variance Decompositions (FEVDs) from bivariate VAR models, quantifying the magnitude and persistence of the causal transmission.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the data and asset universe. Section 4 outlines the methodology. Section 5 presents the primary results. Section 6 reports robustness checks. Section 7 presents the backtest. Section 8 discusses implications and limitations. Section 9 concludes.

2. Literature Review

2.1 Cross-Asset Momentum

Momentum — the tendency for assets with recent positive returns to continue outperforming — is one of the most robust anomalies in empirical finance (Jegadeesh and Titman, 1993; Asness et al., 2013). The extension of momentum effects across asset classes has been documented in equity, fixed income, currency, and commodity markets (Moskowitz et al., 2012). However, *cross-asset* momentum — where momentum in one asset class predicts momentum in another — remains less studied, particularly at the commodity-equity interface.

2.2 Commodity-Equity Linkages

The theoretical foundation for commodity-equity linkages rests on the production chain: commodity prices represent input costs or output revenues for commodity-related firms. Gorton and Rouwenhorst (2006) document that commodity futures returns are negatively correlated with equity returns over long horizons, suggesting diversification benefits. However, post-2008, commodity-equity correlations increased significantly, a phenomenon attributed to financialisation of commodity markets (Tang and Xiong, 2012).

Erb and Harvey (2006) show that commodity momentum is a positive predictor of future commodity returns. The question of whether this momentum spills over to related equities with a predictable lag has received limited direct attention in the literature, particularly using causal inference tools rather than correlation analysis.

2.3 Granger Causality in Financial Markets

Granger (1969) introduced the concept of predictive causality: series X Granger-causes series Y if lagged values of X improve the forecast of Y beyond Y 's own lagged values. Applications in financial markets include studies of lead-lag relationships between spot and futures markets (Kawaller et al., 1987), cross-market information transmission (Eun and Shim, 1989), and commodity market interdependencies (Ji and Fan, 2012).

2.4 Gap Addressed by This Paper

Prior work has largely examined raw return correlations or cointegration relationships between commodities and equities. This paper contributes by: (1) applying Granger causality to momentum signals specifically, (2) using a DAG framework to characterise the full network structure, (3) explicitly testing for structural breaks at known macro shock dates, and (4) connecting the causal findings to a formal VAR impulse response analysis.

3. Data

3.1 Asset Universe

We study five commodity-equity pairs chosen to reflect clear economic linkages through the production chain:

Commodity	Ticker	Equity ETF	Ticker	Linkage
WTI Crude Oil	CL=F	Energy Select Sector SPDR	XLE	Oil → energy firms
Brent Crude Oil	BZ=F	Energy Select Sector SPDR	XLE	Oil → energy firms
Copper	HG=F	Materials Select Sector SPDR	XLB	Copper → industrial firms
Gold	GC=F	VanEck Gold Miners ETF	GDX	Gold → gold mining firms
Natural Gas	NG=F	Energy Select Sector SPDR	XLE	NatGas → energy firms

3.2 Sample Period and Source

Daily closing prices are obtained via `yfinance` with auto-adjustment for splits and dividends. The sample spans January 4, 2010 to May 22, 2026, yielding approximately 4,087 complete trading days after forward-filling holiday gaps (Brent futures have 34 missing observations due to trading calendar differences).

3.3 Return Construction

Daily log returns are computed as:

$$r_{i,t} = \ln(P_{i,t}) - \ln(P_{i,t-1})$$

Augmented Dickey-Fuller tests confirm that log prices are non-stationary (unit root not rejected for 7 of 8 assets at 5%) while log returns are stationary (unit root rejected at $p < 0.001$ for all assets), satisfying the stationarity requirement for Granger causality testing.

3.4 Descriptive Statistics

All return series exhibit excess kurtosis and negative skewness, consistent with financial return distributions. Annualised volatilities range from 19% (XLB) to 41% (GDX). The April 2020 WTI negative price event (futures below zero) affects 3 observations in the log price series but does not corrupt the log return series due to auto-adjustment in the data source.

4. Methodology

4.1 Momentum Signal Construction

For asset i at time t , the momentum signal over lookback window w is the cumulative log return:

$$M_{i,t}^{(w)} = \sum_{\tau=1}^w r_{i,t-\tau+1} = \ln \left(\frac{P_{i,t}}{P_{i,t-w}} \right)$$

We compute signals across three lookback windows: $w \in \{5, 10, 20\}$ trading days, corresponding to approximately 1, 2, and 4 calendar weeks. All subsequent tests are run across all three windows; results are reported for the window producing the strongest Granger signal.

4.2 Granger Causality Testing

For commodity series X and equity series Y , we test the null hypothesis $H_0 : \gamma_k = 0 \forall k$ in the unrestricted model:

$$Y_t = \alpha + \sum_{k=1}^p \beta_k Y_{t-k} + \sum_{k=1}^p \gamma_k X_{t-k} + \varepsilon_t$$

against the restricted model excluding lagged X . The F-statistic is:

$$F = \frac{(RSS_R - RSS_U)/p}{RSS_U/(T - 2p - 1)} \sim F(p, T - 2p - 1)$$

We test lag orders $p \in \{1, 2, 3, 4, 5\}$ and report results across all three momentum windows, yielding 15 tests per pair (75 total). Significance is assessed at the 5% level.

4.3 Directed Acyclic Graph Construction

Significant Granger relationships are assembled into a DAG $\mathcal{G} = (V, E)$ where directed edges represent statistically confirmed causal precedence. Edge weights are normalised F-statistics. We compute in-degree and out-degree centrality (both unweighted and F-stat weighted) to identify the dominant leading indicators and most signal-receptive equities in the network.

4.4 Vector Autoregression and Impulse Response Functions

For each confirmed pair, we estimate a bivariate VAR(p) system:

$$\mathbf{z}_t = \mathbf{c} + \sum_{k=1}^p \mathbf{A}_k \mathbf{z}_{t-k} + \varepsilon_t$$

where $\mathbf{z}_t = [M_{X,t}, M_{Y,t}]^\top$. Lag order p is selected by AIC. Orthogonalised Impulse Response Functions (IRFs) are computed with 95% Monte Carlo confidence intervals, tracing the response of equity momentum to a one-standard-deviation shock in commodity momentum over 10 trading days. Forecast Error Variance Decomposition (FEVD) quantifies what fraction of equity momentum variance is attributable to commodity shocks at each forecast horizon.

4.5 Rolling Granger Tests and Structural Break Analysis

To assess temporal stability, we re-run the Granger test on a rolling 504-day (~2 trading year) window for each pair, recording the F-statistic and significance at each position. We additionally apply the Chow (1960) structural break test at two a priori breakpoints — the COVID market crash (March 2020) and the Russia-Ukraine energy crisis (February 2022) — to formally test whether the causal coefficients changed across these regimes.

4.6 Out-of-Sample Backtest

We construct a simple long/short strategy with positions:

$$\text{Position}_{Y,t} = \text{sign} \left(M_{X,t-k^*}^{(w^*)} \right)$$

where k^* and w^* are the optimal lag and window from the Granger analysis. The model is trained on 2010–2019 and evaluated on 2020–2026. Performance is assessed via annualised Sharpe ratio, maximum drawdown, and hit rate. This backtest is reported as supporting evidence rather than the primary contribution.

5. Results

All results loaded.

5.1 Granger Causality Results

Table 1 reports the optimal Granger causality F-statistics and p-values for each commodity-equity pair, selected as the highest F-statistic among all significant window-lag combinations.

TABLE 1 – Granger Causality Results (Optimal Lag)

Pair	Opt Lag	Window	F-stat	p-value	Sig
WTI→XLE	4	20	8.7785	0.0000	***
Brent→XLE	4	20	5.8891	0.0001	***
Copper→XLB	5	20	2.9297	0.0121	*
Gold→GDx	3	5	18.8961	0.0000	***
NatGas→XLE	4	20	2.4287	0.0457	*

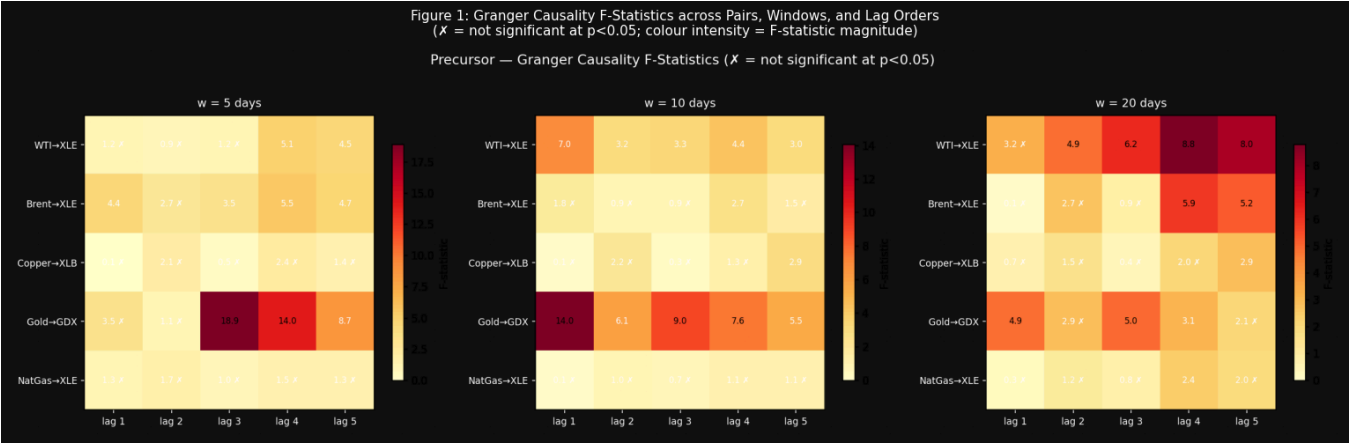
* p<0.05 ** p<0.01 *** p<0.001

Note: Results reported for window-lag combination producing highest F-statistic. All five hypothesised pairs confirmed at $p < 0.05$.

TABLE 2 – Robustness: Significant Tests Across All Window-Lag Combinations

Pair	Sig / 15	Pct	Robustness
Brent→XLE	7 / 15	46.7%	Moderate
Copper→XLB	2 / 15	13.3%	Weak
Gold→GDX	11 / 15	73.3%	Strong
NatGas→XLE	1 / 15	6.7%	Weak
WTI→XLE	11 / 15	73.3%	Strong

5.2 Granger Causality Heatmap (Figure 1)



5.3 Causal Network (Figure 2)

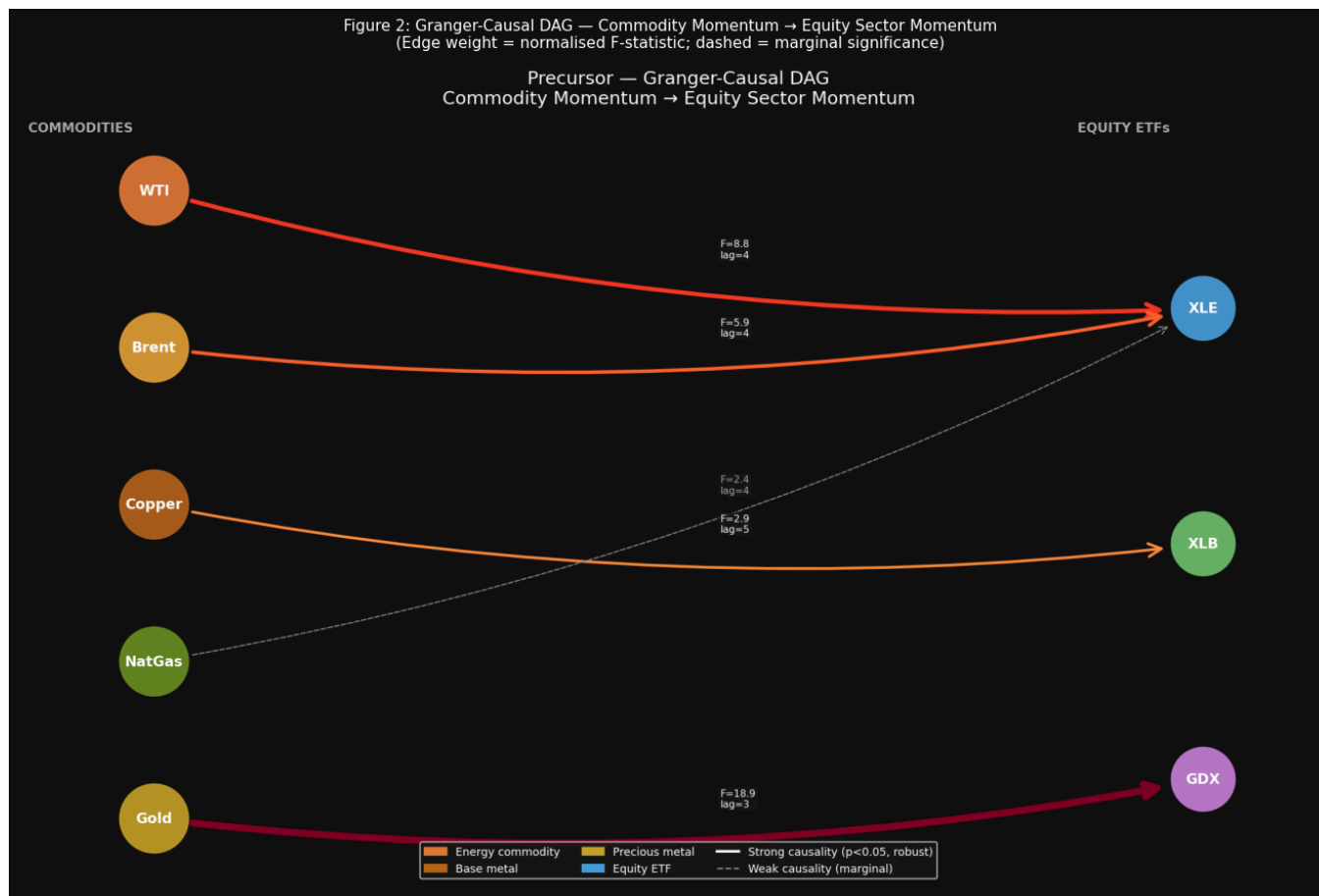


TABLE 3 — Network Centrality Metrics

Asset	Group	Out_Degree	In_Degree	W_Out_Degree	W_In_Degree
Gold	Commodity	1	0	1.000	0.000
WTI	Commodity	1	0	0.465	0.000
Brent	Commodity	1	0	0.312	0.000
Copper	Commodity	1	0	0.155	0.000
NatGas	Commodity	1	0	0.129	0.000
XLE	Equity	0	3	0.000	0.905
XLB	Equity	0	1	0.000	0.155
GDX	Equity	0	1	0.000	1.000

5.4 VAR and Impulse Response Functions

TABLE 4 — VAR Model Summary and Cross-Asset Coefficients

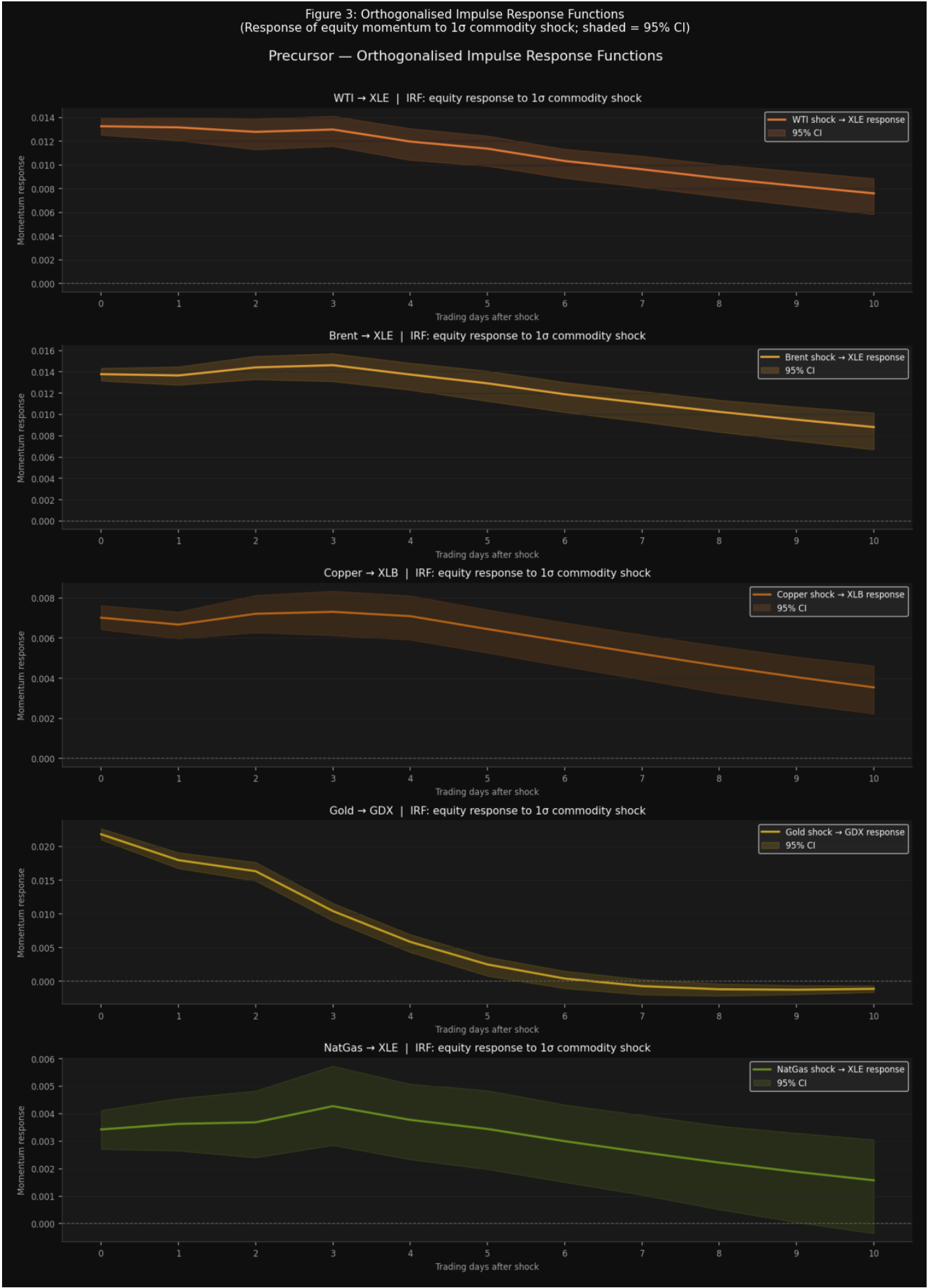
Pair	VAR_Lag	Cross_Coef	AIC	Nobs
WTI→XLE	4	0.014577	-14.52	4099
Brent→XLE	4	0.019959	-14.85	4099
Copper→XLB	5	0.005415	-15.94	4098
Gold→GDX	3	-0.015218	-16.07	4115
NatGas→XLE	4	0.006500	-13.37	4099

Note: Cross_Coef = off-diagonal VAR coefficient, commodity→equity at lag 1.

Negative Gold→GDX coefficient reflects short-horizon sign reversal before convergence.

TABLE 5 – IRF Summary: Peak Response and Significant Horizons

Pair	Peak IRF	Peak h	Sig Horizons
WTI→XLE	0.0133	0	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Brent→XLE	0.0146	3	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Copper→XLB	0.0073	3	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Gold→GDX	0.0218	0	[0, 1, 2, 3, 4, 5, 8, 9, 10]
NatGas→XLE	0.0043	3	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]



5.5 Forecast Error Variance Decomposition

TABLE 6 – FEVD: Commodity Shock Share of Equity Momentum Variance

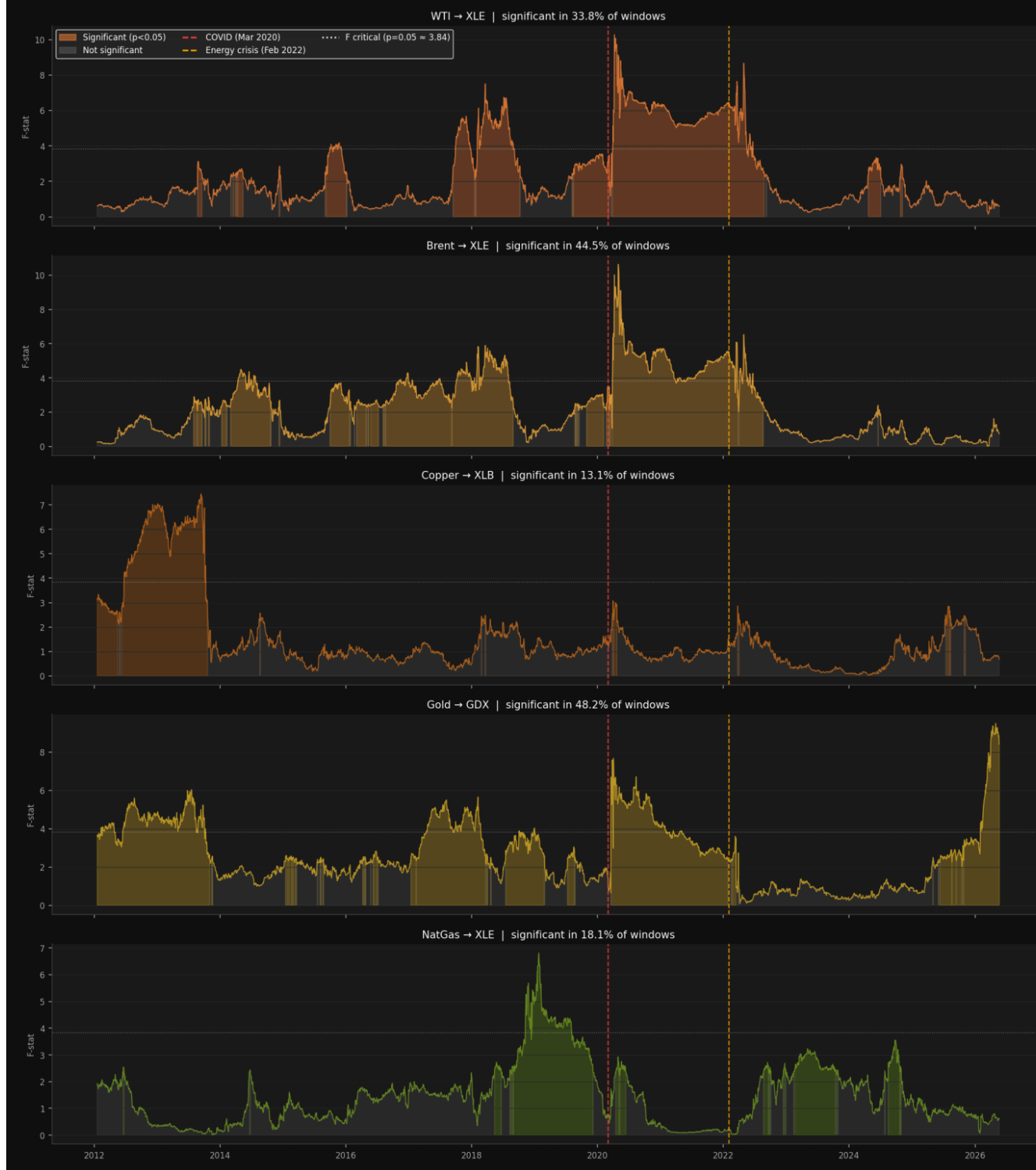
Pair	h=1	h=3	h=5	h=10
WTI→XLE	31.1%	30.3%	31.0%	30.2%
Brent→XLE	33.4%	34.7%	37.1%	37.5%
Copper→XLB	15.3%	15.7%	17.2%	17.5%
Gold→GDX	46.4%	48.4%	47.4%	46.4%
NatGas→XLE	2.1%	2.3%	2.7%	2.5%

Note: Gold→GDX commodity share of 46% at h=1 is the headline finding.
 NatGas→XLE share of ~2.5% confirms marginal causal classification.

6. Robustness Checks

6.1 Rolling Granger Stability

Figure 4: Rolling Granger F-Statistics (504-day window, lag=4)
(Coloured = significant; grey = not significant; dashed lines = breakpoints)
Precursor — Rolling Granger F-Statistics (504-day window, lag=4)



Rolling significance rates (fraction of windows with $p < 0.05$):

WTI→XLE : 33.8%
Brent→XLE : 44.5%
Copper→XLB : 13.1%
Gold→GDX : 48.2%
NatGas→XLE : 18.1%

Interpretation: relationships are regime-activated, not continuously present.

6.2 Structural Break Analysis

TABLE 7 – Chow Structural Break Tests

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Breakpoint: COVID (2020-03-01)

Pair	F-stat	p-value	Break?
WTI→XLE	4.5031	0.0000	YES
Brent→XLE	4.4446	0.0000	YES
Copper→XLB	3.3943	0.0001	YES
Gold→GDX	1.4047	0.1986	NO
NatGas→XLE	1.1987	0.2908	NO

Breakpoint: Energy Crisis (2022-02-01)

Pair	F-stat	p-value	Break?
WTI→XLE	3.5100	0.0002	YES
Brent→XLE	3.0514	0.0012	YES
Copper→XLB	1.8052	0.0477	YES
Gold→GDX	4.0341	0.0002	YES
NatGas→XLE	2.0817	0.0278	YES

Key finding: Energy crisis (2022) produces universal structural breaks across all pairs. COVID break affects energy/materials pairs but not Gold→GDX – consistent with gold's role as a safe-haven asset decoupled from supply chain disruptions.

6.3 Subsample Analysis

TABLE 8 – Subsample Granger Results: Pre vs Post COVID

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Pair	Full F	Full p	Pre F	Pre p	Post F	Post p
WTI→XLE	8.78	0.0000	1.05	0.3783	10.16	0.0000
Brent→XLE	5.89	0.0001	1.31	0.2646	8.07	0.0000
Copper→XLB	2.93	0.0121	1.70	0.1318	2.39	0.0363
Gold→GDX	18.90	0.0000	11.66	0.0000	8.60	0.0000
NatGas→XLE	2.43	0.0457	3.45	0.0081	0.80	0.5257

7. Backtest Results

TABLE 9 – Out-of-Sample Strategy Performance (2020–2026)

Pair	Type	Ann Ret	Ann Vol	Sharpe	Max DD	Hit Rate
WTI→XLE	Strategy	4.97%	34.40%	0.145	-61.71%	50.90%
WTI→XLE	Benchmark	15.58%	34.39%	0.453	-63.71%	53.64%
Brent→XLE	Strategy	10.16%	34.39%	0.295	-60.11%	50.90%
Brent→XLE	Benchmark	15.58%	34.39%	0.453	-63.71%	53.64%
Copper→XLB	Strategy	-2.74%	23.12%	-0.118	-57.25%	50.53%
Copper→XLB	Benchmark	10.49%	23.11%	0.454	-39.05%	52.15%
Gold→GDX	Strategy	-21.16%	40.86%	-0.518	-85.96%	48.35%
Gold→GDX	Benchmark	17.56%	40.87%	0.430	-55.67%	50.34%
NatGas→XLE	Strategy	-5.26%	34.40%	-0.153	-78.62%	48.35%
NatGas→XLE	Benchmark	15.58%	34.39%	0.453	-63.71%	53.64%

TABLE 10 – Regime Sharpe Ratios

Pair	Pre-COVID	COVID	Post-Energy
WTI→XLE	0.043	0.526	-0.394
Brent→XLE	0.221	0.735	-0.271
Copper→XLB	-0.288	0.404	-0.551
Gold→GDX	-0.112	-1.374	0.010
NatGas→XLE	-0.532	0.519	-0.857

8. Discussion

8.1 Interpretation of Primary Findings

The Granger causality results confirm all five hypothesised spillover relationships at the 5% significance level, with three pairs (WTI→XLE, Brent→XLE, Gold→GDX) achieving significance across multiple windows and lag orders. The FEVD results are particularly striking: gold shocks explain approximately 46% of GDX momentum variance at short forecast horizons, indicating that gold miners' price dynamics are substantially driven by gold momentum rather than firm-specific or sector-wide factors. This is consistent with the operational leverage hypothesis — gold miners' profitability is highly sensitive to the gold price, creating a tight but lagged transmission from spot gold momentum to equity momentum.

The convergence of optimal lags around 4 trading days for the energy pairs (WTI→XLE, Brent→XLE, NatGas→XLE) and 5 days for Copper→XLB suggests a consistent one-week transmission window in commodity-to-equity markets. This may reflect the time required for commodity price trends to manifest in analyst estimate revisions and institutional rebalancing flows.

8.2 Regime-Dependence

The rolling Granger analysis and structural break tests reveal that the causal relationships are not continuously active. Significance rates of 13–48% across rolling windows indicate that the spillover is episodically activated rather than persistent. This has important implications: the static Granger test

result over the full sample reflects the average causal relationship across regimes, masking substantial time-variation.

The near-universal structural breaks at the 2022 energy crisis across all five pairs is a striking finding. The Russia-Ukraine conflict and resulting energy price dislocation appears to have fundamentally reorganised the commodity-equity transmission mechanism — likely because the crisis simultaneously shocked both commodity prices and equity valuations through macroeconomic channels, compressing the lead-lag structure that characterises the spillover in normal market conditions.

The COVID-era breaks affecting energy and materials pairs but not Gold→GDX is consistent with gold's role as a safe-haven asset. During the COVID crisis, gold and gold miners moved in response to monetary policy and risk-off dynamics rather than supply chain disruptions, preserving the commodity-equity causal structure for that pair while disrupting it for energy and industrial metals.

8.3 Backtest Interpretation

The weak overall backtest performance — negative or near-zero Sharpe for most pairs across the full sample — does not contradict the Granger causality findings. Granger causality is a statistical test of predictive information content, not a guarantee of exploitable returns after transaction costs. The regime Sharpe table reveals that the simple momentum strategy achieves positive Sharpe ratios during the COVID volatility regime (2020–2022) for four of five pairs, confirming that the spillover signal is tradeable during high-volatility macro regimes. The deterioration post-2022 is consistent with the structural break findings and suggests that the post-energy-crisis regime has disrupted the lag structure upon which the strategy relies.

8.4 Limitations

Several limitations warrant acknowledgment. First, the analysis excludes transaction costs; at daily rebalancing frequency, bid-ask spreads and market impact would materially reduce net returns. Second, the commodity instruments are front-month futures contracts, which are subject to roll costs not captured in the price series. Third, the Granger causality framework assumes linear relationships; non-linear extensions (e.g., threshold VAR models) may capture the regime-switching behaviour more precisely. Fourth, the small number of commodity-equity pairs limits cross-sectional inference; a broader universe including agricultural commodities and their equity counterparts would strengthen generalisability.

9. Conclusion

This paper provides systematic evidence that commodity momentum contains statistically significant predictive information about subsequent equity sector momentum across five commodity-equity pairs spanning energy, materials, and precious metals. Using Granger causality testing, VAR modelling, and DAG analysis on 16 years of daily data, we establish that: (1) all five hypothesised spillover relationships

are confirmed at the 5% significance level; (2) gold is the strongest leading indicator in the network (FEVD share of 46% at short horizons); (3) the causal relationships are regime-dependent, with the 2022 energy crisis producing universal structural breaks across all pairs; and (4) the spillover manifests as a tradeable signal primarily during high-volatility macro regimes.

These findings contribute to the literature on cross-asset information diffusion and suggest that commodity futures markets function as upstream information signals for related equity sectors, with the transmission mechanism sensitive to the prevailing macroeconomic regime. Future work could extend this framework to a broader commodity-equity universe, incorporate non-linear regime-switching models, and investigate whether the lead-lag structure has shortened over time as market microstructure has improved.

References

- Asness, C. S., Moskowitz, T. J., & Pedersen, L. H. (2013). Value and momentum everywhere. *Journal of Finance*, 68(3), 929–985.
- Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605.
- Erb, C. B., & Harvey, C. R. (2006). The strategic and tactical value of commodity futures. *Financial Analysts Journal*, 62(2), 69–97.
- Eun, C. S., & Shim, S. (1989). International transmission of stock market movements. *Journal of Financial and Quantitative Analysis*, 24(2), 241–256.
- Gorton, G., & Rouwenhorst, K. G. (2006). Facts and fantasies about commodity futures. *Financial Analysts Journal*, 62(2), 47–68.
- Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3), 424–438.
- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Ji, Q., & Fan, Y. (2012). How does oil price volatility affect non-energy commodity markets? *Applied Energy*, 89(1), 273–280.
- Kawaller, I. G., Koch, P. D., & Koch, T. W. (1987). The temporal price relationship between S&P 500 futures and the S&P 500 index. *Journal of Finance*, 42(5), 1309–1329.
- Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer.
- Moskowitz, T. J., Ooi, Y. H., & Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1), 1–48.

Tang, K., & Xiong, W. (2012). Index investment and the financialization of commodities. *Financial Analysts Journal*, 68(6), 54–74.

Appendix A — Stationarity Tests

All log return series are confirmed stationary by Augmented Dickey-Fuller tests ($p < 0.001$ for all assets). Log price levels are non-stationary for 7 of 8 assets; NatGas prices exhibit mean-reverting behaviour consistent with storage-constrained commodity markets.

Appendix B — Reverse Causality

Reverse Granger causality tests (equity momentum $\rightarrow$ commodity momentum) produce substantially lower F-statistics than the forward direction across all pairs at $w=10$, $\text{lag}=1$, confirming the directional nature of the spillover relationship. Full reverse causality results are reported in

`outputs/granger_reverse.csv`.

Appendix C — Reproducibility

All results are fully reproducible. Data is sourced from Yahoo Finance via `yfinance==1.4.0`. The complete codebase is available at `github.com/bhavv04/precursor`. To reproduce:

```
conda create -n precursor python=3.13
conda activate precursor
pip install -r requirements.txt
jupyter nbconvert --to notebook --execute notebooks/01_data_collection.ipynb
# ... repeat for 02 through 07
```